



Chapter 2: Construction Planning and Scheduling

2.1 Construction Planning

a) Definition:

- Planning is the **mental process** deciding about the future line of action
- It is the **course of action** to achieve the desired result taking in to consideration the **present needs** and **future requirement**
- Deciding about **what** to do, **when** to do, **who** will do and **where** to do is planning
- Various **alternative methods** of **execution of work** are studied at the planning stage and **decisions are taken for each part** of work



Planning

PLANNING PROCESS AND ITS TYPES

EVERSHINE/SUSITRA/T2/M2/



WHAT IS CONSTRUCTION PLANNING ?

- Construction planning is the first stage of construction management, the discipline of taking a construction project from conception to completion.
- It includes components such as

- *Scheduling*
- *Organising*
- *Staffing*
- *Directing*
- *Monitoring*



Reference : <https://www.smart-sheet.com/construction-project-planning>



Planning

b) Objectives of Planning:

- Proper **design** of each element
- Proper selection of **equipment**
- Proper arrangement and **repair of equipment**
- **Procurement** of required materials in advance
- Ensure employment of **trained and experienced staff**
- To provide **welfare schemes** for staffs
- To provide **incentives** to good worker
- To arrange constant **flow of fund**
- To provide proper **safety measures**
- To provide proper arrangement of **communication**



Planning

c) Principles of Planning:

- Plan should be **readily understandable** (i.e. simple language & able to convey correct message)
- It should be **realistic, flexible** and **comprehensive**
- Can be used as the basis for **monitoring and controlling** the project





Planning



Principles of Planning

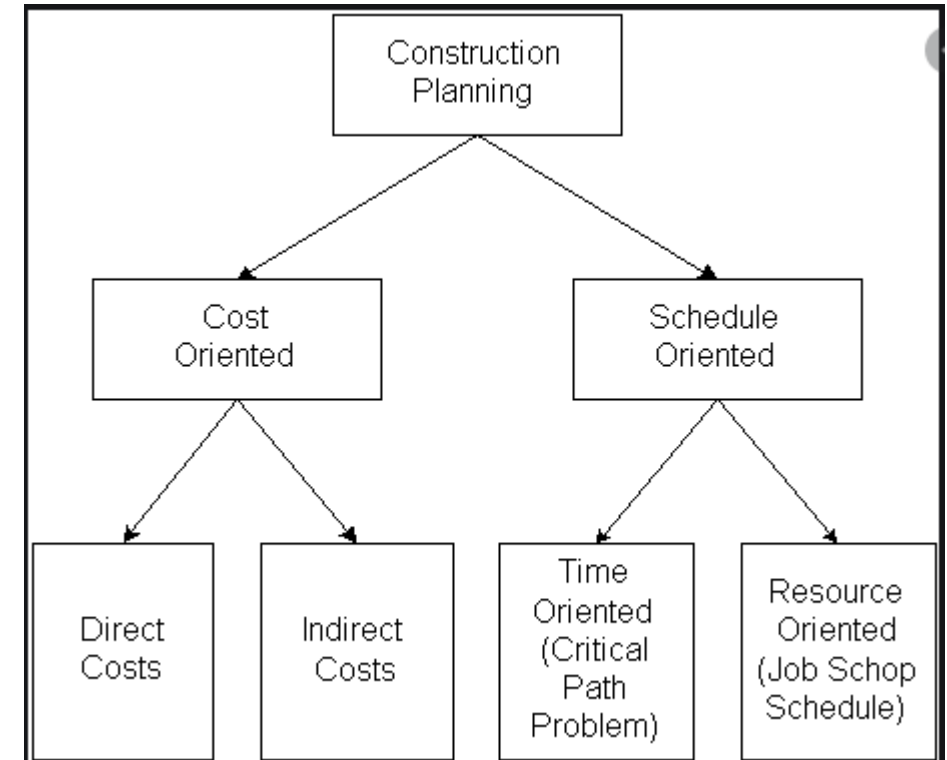
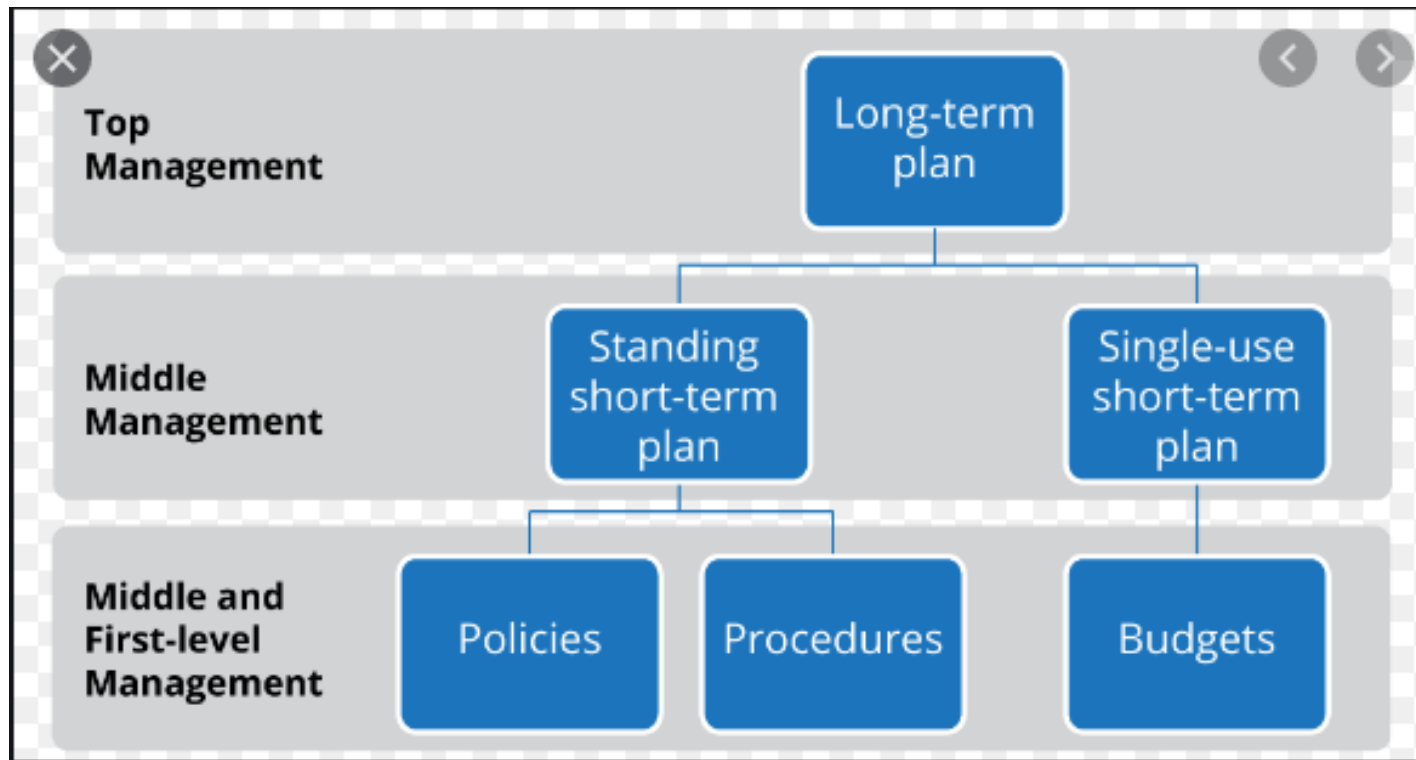


- These are the guidelines for planning the buildings to suit their functional requirements.
- **Various principles of planning:-**
 1. Aspect
 2. Prospect
 3. Privacy
 4. Roominess
 5. Grouping
 6. Circulation
 7. Elegance
 8. Sanitation
 9. Orientation
 10. Economy



Planning

d) Types of Plan:



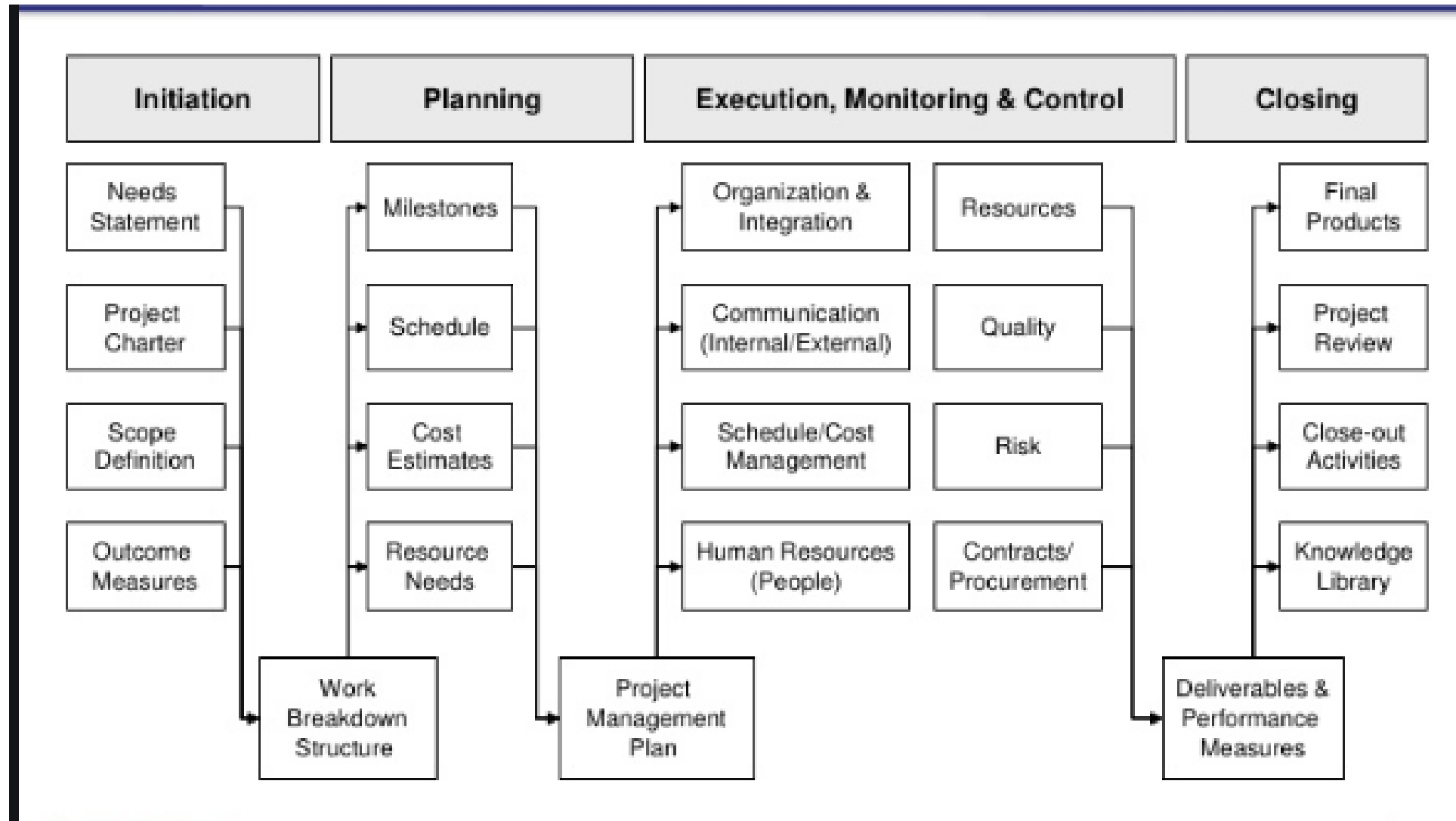


Types of Plan





Project Management Activities





2.2 Steps and Stages of Planning





Steps in Planning

- **Identify the problems** to be encountered and obtain the necessary **information useful in execution**
- Obtain **alternative ways** of execution work and select the optimum plan
- **Fix time of start** of work
- Decide **quantity and types of resources**
- Estimate project **duration**
- Estimate **financial help**
- Evaluation of **effectiveness of plan**



Steps in Planning





Stages of Planning

1) Pre-Planning:

- **Objectives** are clearly **spelt out**
- **General framework** of the project is to be formulated
- **Justification** for the construction **of the project** is to be laid down
- **Cost benefit analysis** of the project is to be carried out
- Cost analysis for all **alternative sites** is to be done

2) Detailed Planning:

- Preparation of detailed **design**
- Preparation of drawings and **specifications**
- Working out **quantities** of **materials**
- Breaking the project in to different **alternatives**
- Detailing the **sequence of operation** of activities & allocating duration



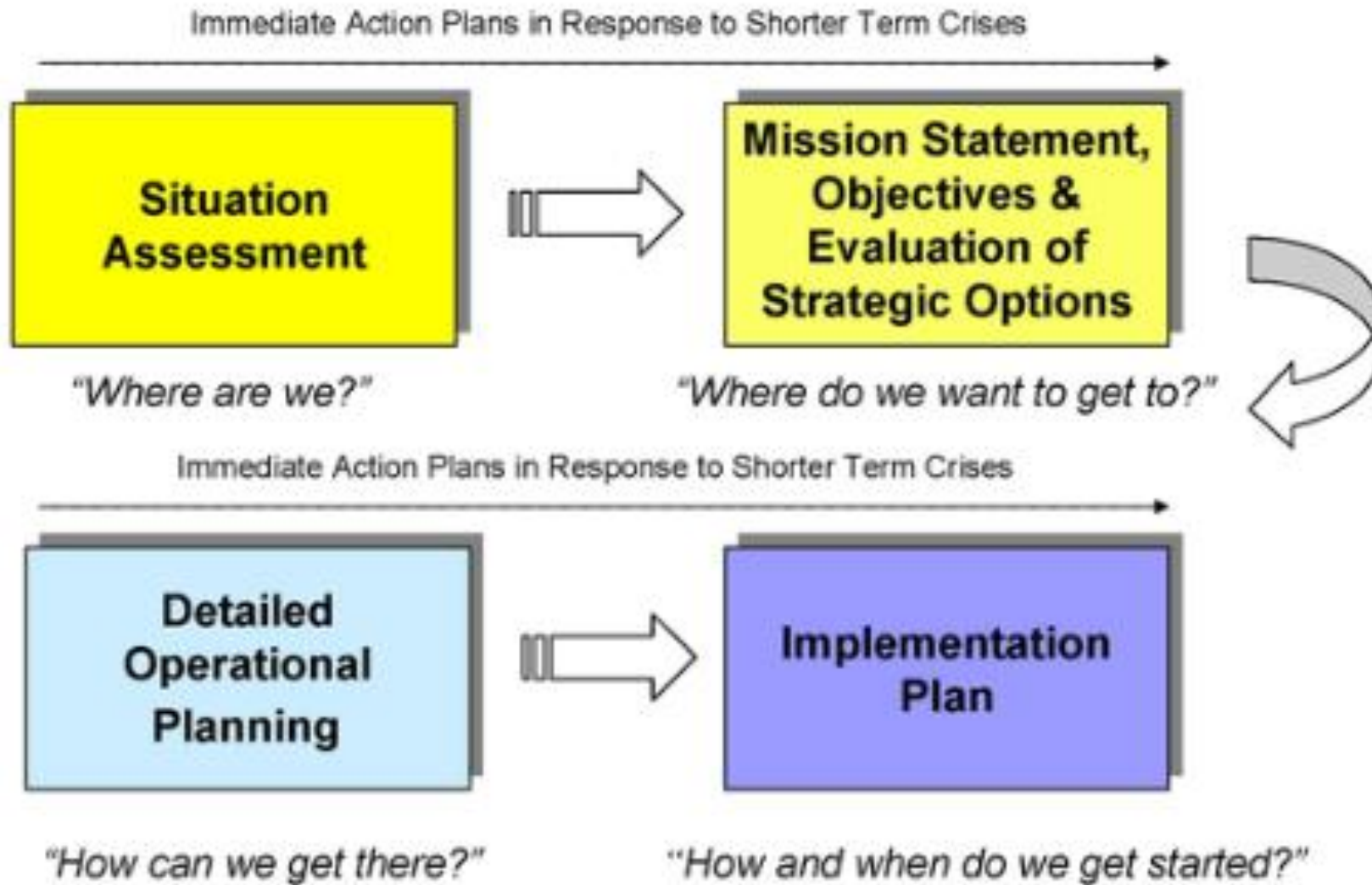
Stages of Planning

3) Monitoring and Control:

- **Progress** of construction is **monitored as per proposal**
- **Updating the schedule** as per progress
- Preparing the **revised forecast**



Stages of Planning





2.3 Planning by Client and Contractor in Different Stages

A) Planning by Client: Client may be an individual or group of individuals or Government and it's agencies.





Role of Stakeholders in Construction

Role	Description	Actors
Customer	Ordering built product, representing client's interests and supplying design information not controlled by the contractor	Client, Project Manager, Architect, QS, Structural / M&E Engineers
Performer	Planning construction as part of tender offer to supply built product	Principal contractor's planning team responsible to senior management
Supplier	Providing contractor with information on supply chain delivery capabilities	Trusted package contractors and specialist material suppliers
Constrainer	Controllers of the site and its environment who have the power to block or restrict construction operations	Client and/or client's landlord Utilities, adjoining owners, police, planning/highway authorities



Level of Planning

Level	Level of planning	Undertaken by
1	project planning	client/project manager
2	pre-tender planning	tendering contractors
3	pre-contract planning	main contractor
4	contract planning	main contractor and subcontractors



A) Planning by Contractor

a) Pre-tender Stage Planning:

FACTORS TO CONSIDER:

- No. of trade enquiries – competition
- Policy of sharing – good relations
- Dates of sub-contracts
- Compliance with bill of quantities and specifications
- Consistency of unit rates throughout quotation
- Rates realistic in relation to that of competitions
- Preliminaries and attendances included?
- Compliance with main contract (payment conditions/retention)
- Onerous (burdensome) conditions
- Quality and extent of supervision
- Competence and mastery of speciality
- Degree to which dependent on further subcontracting
- Availability and condition of equipment
- Economic and financial stability
- Labour policies and practices
- Expecting capability
- Previous experiences that you and other contractors have had with him (the subcontractor)



Work Activities of Contractor

Work Activities (Contractor)
Site Assessment
Award of Contract
Endorsement of Basic Design and Documents
Performing of the Detail Design
Performing the procurement
Fabrication of Materials
Load-Out and Transportation
Launch and Installation
Commissioning



b) Contract Stage Planning

- After **acceptance of tender**, the contractor further needs **intensive planning**
- Pre-tender stage planning work is fully utilized for this stage also

Steps for Contract stage planning:

- **Study of alternative methods** of construction to choose the most economical one
- **Working out the quantities** of materials, **sources** of supply and **comparative cost** from different source
- Working out of construction **labor, supervisory and managerial staff**
- Selection of **human resource** and their recruitment
- Working out the **types and quantities of machinery** and equipment



b) Contract Stage Planning

- **Repair and maintenance scheme** of equipment
- Planning for location of **camp office, service road layout** and **other facilities**
- Planning and arranging of **transport of labor**
- To study the **interdependency of** different items of **work**
- To finalize the **work program of each item** of work



2.4 Preparing Schedule

a) Definition of Schedule

- It is a **graphical representation** which shows the starting and completing dates of each activity & their sequential relationship
- It is the **time table** for execution of a project

b) Use of scheduling

- **Quantity** of work involved, labor, materials and money required in each stage **can be determined**
- **Actual progress** of can be checked
- Project can be carried out in a **systematic manner**



Scheduling

c) Advantages of scheduling

- PREDICT-ABILITY :- A scheduling system makes everyone's job easier by adding predictability to the AIS environment
- To superiors:- it provides a means of holding down costs through better use of personnel and equipment.
- Effective use of all AIS resources
- Increased throughput
- Decreased turnaround time
- User deadlines met
- Improved communications with users
- Avoidance of overloading and underuse of resources
- Job delays more readily apparent





Scheduling

d) Preparation of Construction Schedule

- Project is divided in to small measurable units of work so that time, budget and other resources can be allocated (i.e. preparation of **WBS**)
- Study of **logical sequence** of each items
- **Estimation of quantities of work** involved in each operation
- **Estimation of time** required for each operation

e) Classification of Scheduling

- **Construction schedule:** It is the **roaster prepared** for the construction of different activities of work for the completion of project.
- **Material schedule:** It should be developed at site at least **one week earlier than it's use.**



Scheduling

- **Labor schedule:** The aim of this schedule is to divide the number of skilled and unskilled labor.
- **Equipment schedule:** To decide type, number and dates on which a particular equipment will be needed. The main aim is to have optimum use of equipment.
- **Financial schedule:** It's aim is to find out the amount of money needed after completion of different event.
- **Control schedule:** At due date, the in charge of the project has to submit the work progress to the head office.
- **Organizational schedule:** It is the Organizational Breakdown Structure
- **Summary schedule:** It provides the out put of the project



Scheduling

f) **Method of Scheduling**

- Bar Chart / Gantt Chart
 - Milestone Chart (Modified Bar Chart)
 - Net Work Analysis (AOA - Activity on Arrow; AON – Activity on Node)
- a) CPM
 - b) PERT



2.5 Time cost Trade Off

Definition:

➤ Time-Cost Trade Off analysis involves **accelerated activity durations** that are obtained **by allocating more resources**, and **lead to shorter project duration and lower indirect cost** at the **expense of higher direct cost**

- Project ***duration can be reduced*** by assigning more resources to project activities.
- However, doing this ***increases project cost***.
- Decision is based on analysis of ***trade-off between time and cost***.
- ***Project crashing*** is a method for shortening project duration by reducing one or more ***critical activities*** to a time less than normal activity time.



Objective of Time Cost Trade Off

- The **objective** of Time Cost Trade Off is to **reduce** the **original project duration**, determined from critical path analysis to **meet a specific dead line**, with the **least cost of the project**.
- **Project duration can be reduced** by adjusting **overlaps between critical activities** or by **reducing the duration of critical activities**.
- While reducing the duration of critical activities, **direct cost** (related the labor cost) **increases** because of extra payment for use of seasonal labors / overtime work etc. but **indirect cost decreases** because of reduction of project duration.
- After doing **such trials**, **total cost** of the project (direct cost + indirect cost) decreases up to **certain limit** and **starts to increase**. While doing many trials, we will get minimum total cost. The project duration corresponding to **minimum total project cost** is **optimum project duration**.
- The **aim** of time cost trade off is **to obtain minimum** possible project **cost**, **optimum project duration** and **corrected schedule** as per optimum project duration.



Terminology related Time Cost Trade Off

- Normal cost
- Normal duration
- Crash cost
- Crash duration
- Cost slope
- Critical path
- Critical activities
- Direct cost
- Indirect cost
- Total cost
- Optimum project duration
- Minimum total cost



Time Cost Trade Off

- Time-Cost Trade-off Model
 - A model that develops the relationship between direct project costs, indirect costs, and time to complete the project by minimizing the sum of direct and indirect costs.
- Crash Costs
 - The additional costs of an activity when time to complete it is shortened.
- Assumptions – **expediting** a project or **lengthening** a project both add to the costs!



Reasons to Reduce Project Durations

1. To realize incentive pay
2. To fit within the contractually required time (*influences Bid price*).
3. To avoid paying Liquidated Damages/Damage for delay causes (*or avoid damaging the company's relationship*).
4. To complete a project early & move on to another project (*Although the Job-specify profit may reduce, company profile may increase*).
5. To complete a project when weather conditions make it less expensive (Avoid temporary Heating, avoid completing site work during raining season).



Direct and Indirect Cost of the Project

Direct Costs:

all costs that are directly associated with the project.

- subcontractors
- hired labor
- materials
- supplies
- tools and equipments
- bonds and permit

Indirect Costs:

costs that cannot be tied to a specific project and need to be allocated to projects.

- compensation and salary of supervisors or other workers that involve in several projects, include benefits and payroll taxes
- general liability insurance
- vehicle repair and depreciation
- maintenance
- subscriptions, memberships fee



Time Cost Trade Off Model

- Develops the relationship between direct costs, indirect costs, and time to complete the project by minimizing the sum of direct and indirect costs.
- **Crash – shortening activity duration**
- **Crash Costs**
 - Additional costs of an activity when time to complete it is shortened.
 - Crashing costs increase as project duration decreases
 - Indirect costs increase as project duration increases



Steps to be followed for Time-Cost Trade Off

- Prepare CPM network and obtain project duration.
- Calculate floats and identify critical activities and critical path/paths.
- Calculate the cost slope of all activities.
- Choose the one critical activity (among the activities) having minimum cost slope and reduce the duration of it by one unit of time (i.e. day or week as per given).
- If we choose more unit at one step then there may chance of being the another path also critical.
- In such situation, we have to add the cost slope of both parallel activities and it must be compared with the cost slope of other activities to get least cost slope for getting direct cost.

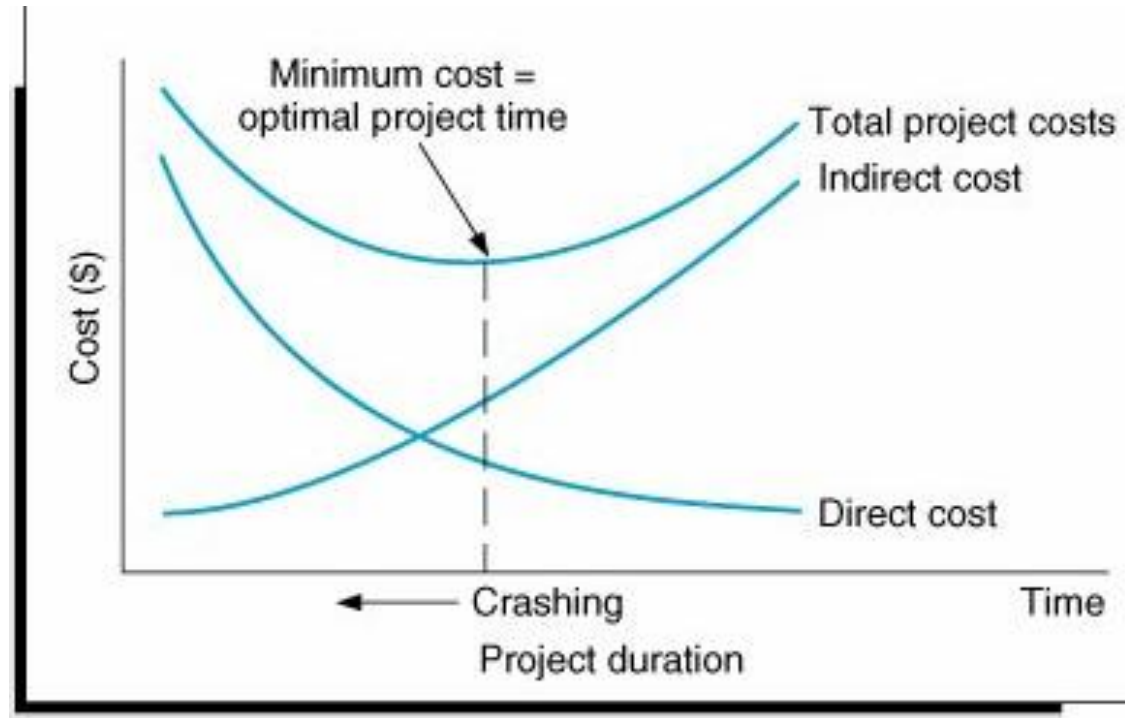


Steps to be followed for Time-Cost Trade Off

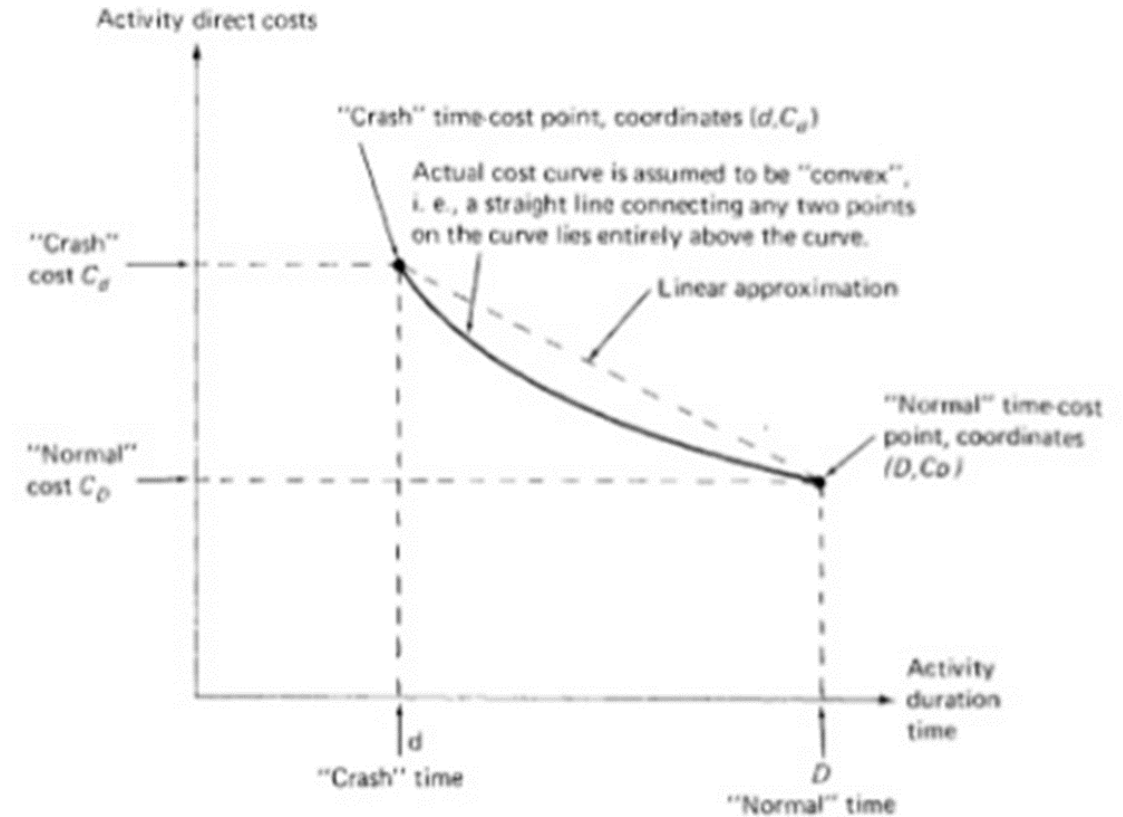
- We have to calculate the indirect cost of the project for each step using the available project duration after crashing the activity/activities since indirect cost (overhead cost) is given.
- Obtain total project cost (direct cost + indirect cost) in each step.
- Up to certain limit, total project cost decreases and becomes minimum then increases.
- Project duration corresponding to minimum total project cost is the optimum project duration.
- Prepare the revised schedule having optimum duration which contains crashed activities.
- Plot the graph for direct cost, indirect cost and total cost.



Time-Cost Trade Off Curve



Balancing (indirect) overhead and direct cost

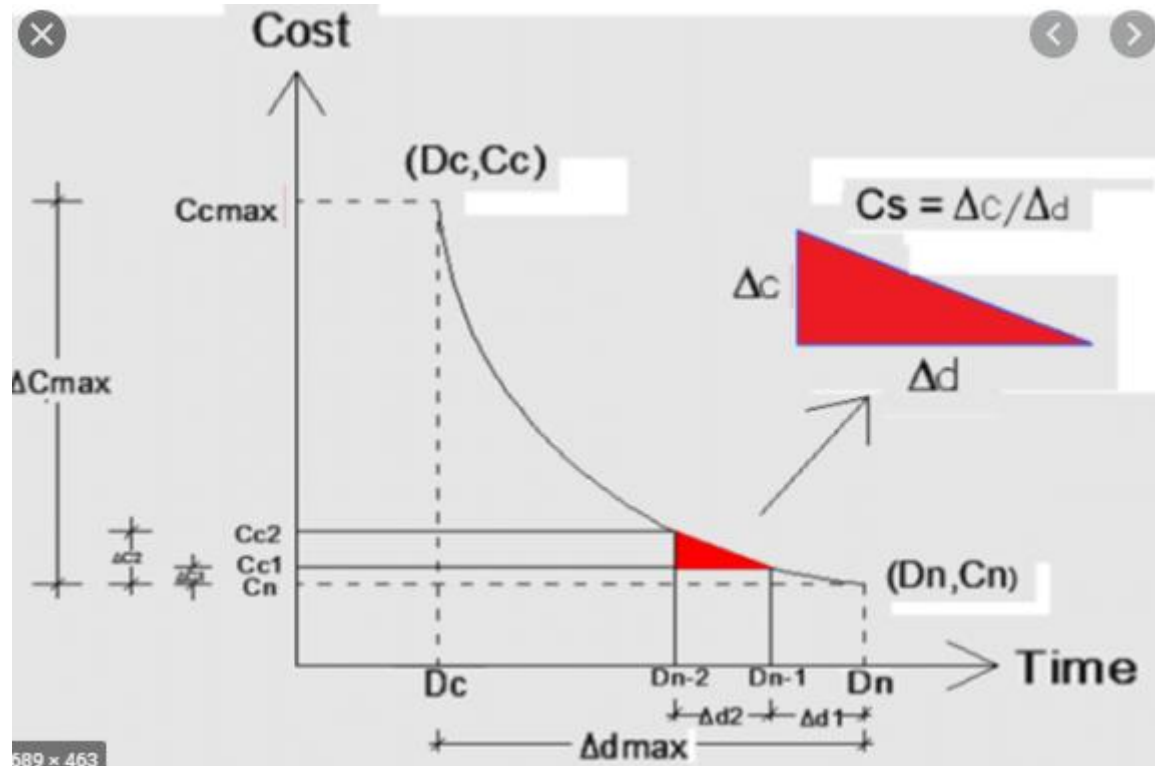


Activity Time Cost Trade Off

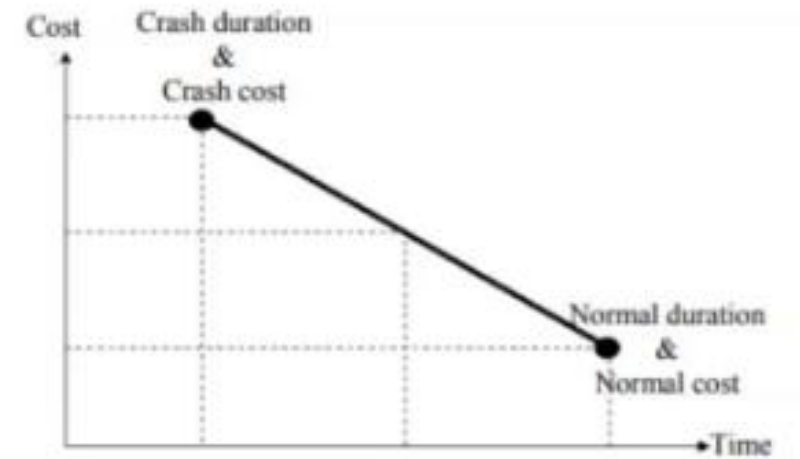


Time Cost Trade Off and Cost Slope for an Activity

By MRD



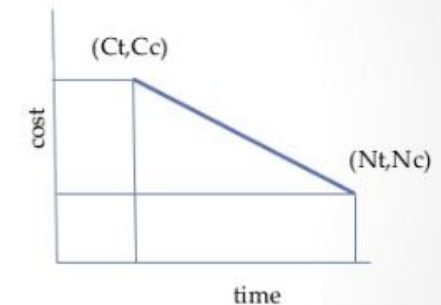
Tim-Cost Trade off and Cost Slope for an Activity



Activity Time-Cost Relationship

Nt: Normal time
Nc: Crashing time
Ct: Cost of activity in Nt.
Cc: Cost of activity in Nc.

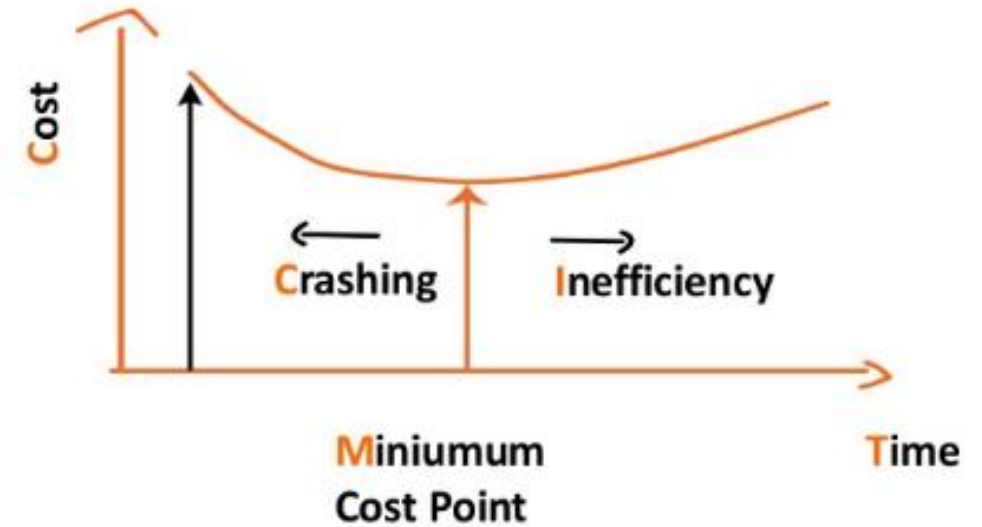
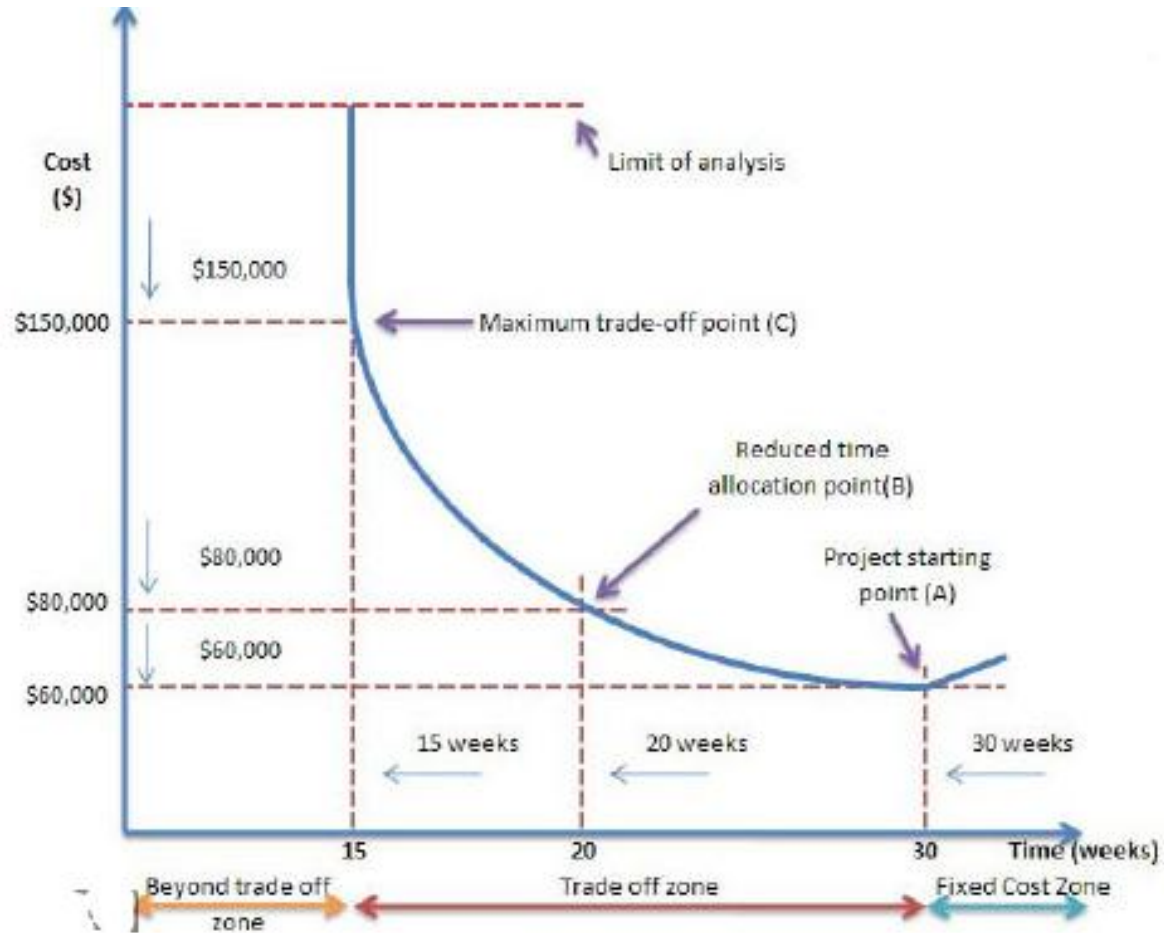
$$\text{Cost Slope} = \frac{Cc - Nc}{Nt - Ct}$$



Cost Slope



Time-Cost Trade Off Curve

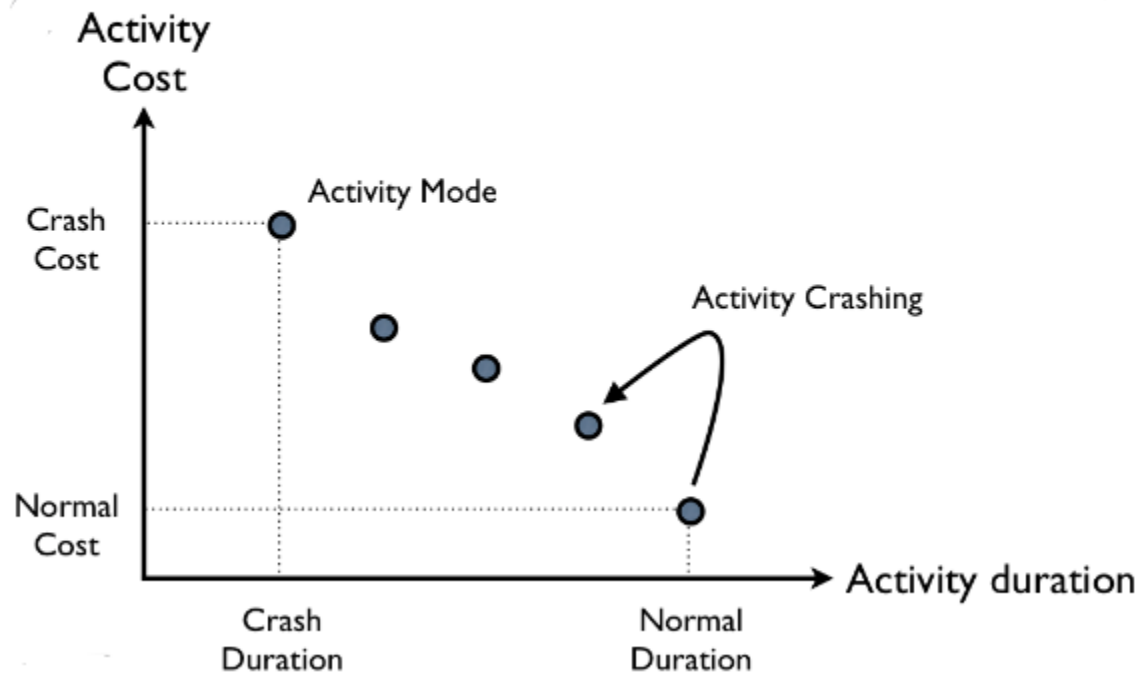


Time-cost trade off

Project cost definition



Time cost Trade off



Time cost trade off in a project schedule

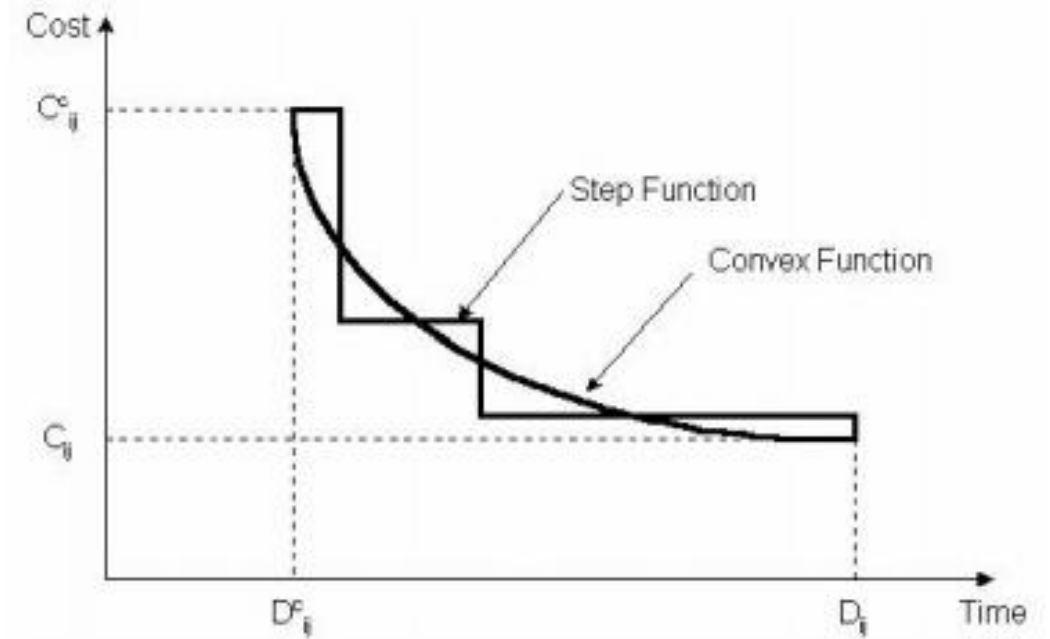
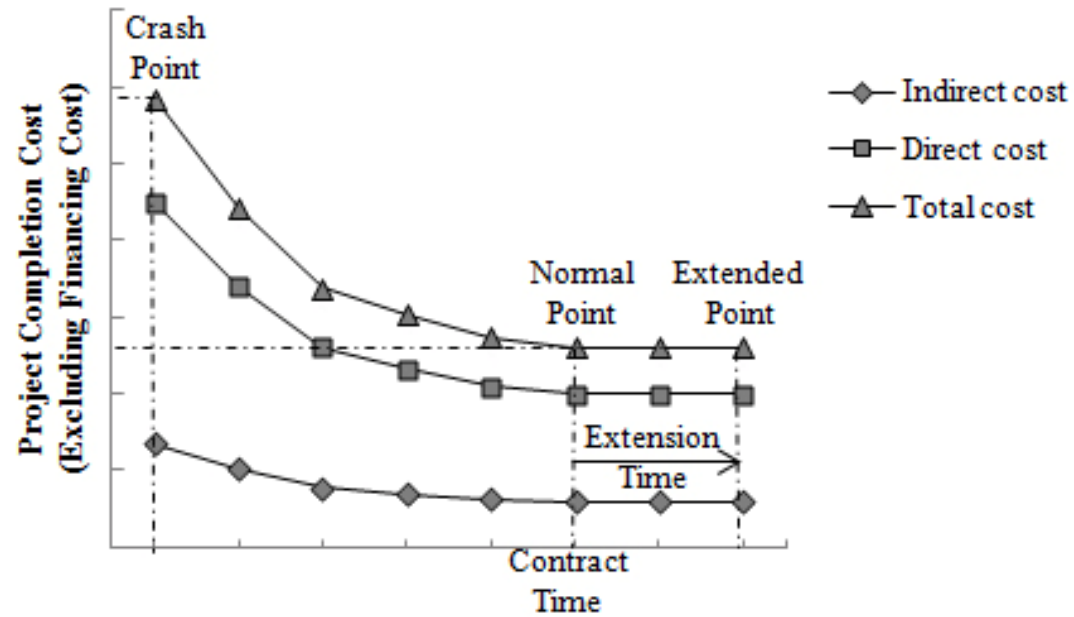


Illustration of Non Linear time cost trade off for an activity



Time Cost Trade Off



Project Completion time

Procedure for crashing

- Crash the project one period at a time
- Only an activity on the critical path
- Crash the least expensive activity
- Multiple critical paths: find the sum of crashing the least expensive activity on each critical path



Tutorial 1. (Time Cost Trade Off)

Consider project time and cost data in Table 1. The indirect cost of the project is \$200 per day.

Table 1. Normal and crash times/costs for the project (*Note:* Times are in days)

Task	Immediate Predecessors	Normal Time (days)	Crash Time (days)	Normal Cost (\$)	Crash Cost (\$)
A	—	6	6	200	200
B	A	10	4	600	1,000
C	A	12	9	625	1,000
D	B	6	5	700	800
E	B	9	7	200	500
F	C, D	9	5	400	840
G	E	14	10	1,000	1,440
H	E, F	10	8	1,100	1,460

Obtain minimum cost of the project using time cost trade off and optimum duration. Also prepare the least cost schedule.



Tutorial 2. (Time Cost Trade Off)

Consider the project time cost data in table 2. The indirect cost of the project is \$400 per week. Prepare the least cost schedule.

Activity	Activity Predecessor	Activity Time (weeks)		Activity Cost (\$)	
		Normal	Crash	Normal	Crash
a	—	16	8	2,000	4,400
b	—	14	9	1,000	1,800
c	a	8	6	500	700
d	a	5	4	600	1,300
e	b	4	2	1,500	3,000
f	b	6	4	800	1,600
g	c	10	7	3,000	4,500
h	d, e	15	10	5,000	8,000



Tutorial 3. (Time Cost Trade Off)

The following table provides the information necessary to construct a project network and project crash data:

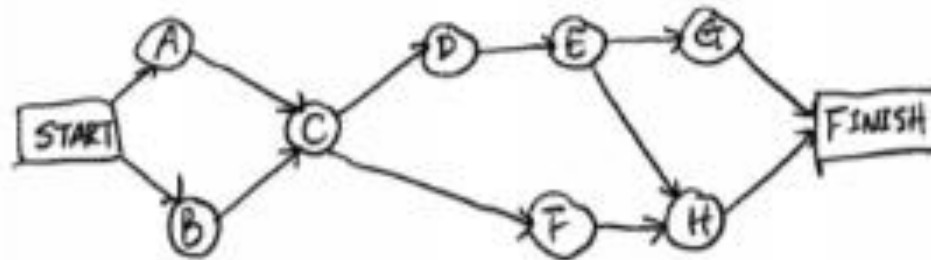
Activity	(i, j)	Activity Predecessor	Activity Time (weeks)		Activity Cost	
			Normal	Crash	Normal	Crash
a	(1, 2)	—	16	8	\$2,000	\$4,400
b	(1, 3)	—	14	9	1,000	1,800
c	(2, 4)	a	8	6	500	700
d	(2, 5)	a	5	4	600	1,300
e	(3, 5)	b	4	2	1,500	3,000
f	(3, 6)	b	6	4	800	1,600
g	(4, 6)	c	10	7	3,000	4,500
h	(5, 6)	d, e	15	10	5,000	8,000

Overhead per week is \$450. Obtain the least cost schedule and optimum duration of it.



Tutorial 4. (Time Cost Trade Off)

The following table provides the data related to the normal and crash time and cost of a construction project. A network diagram of it is also given. Compute the least cost schedule. The indirect cost per week is \$7500.

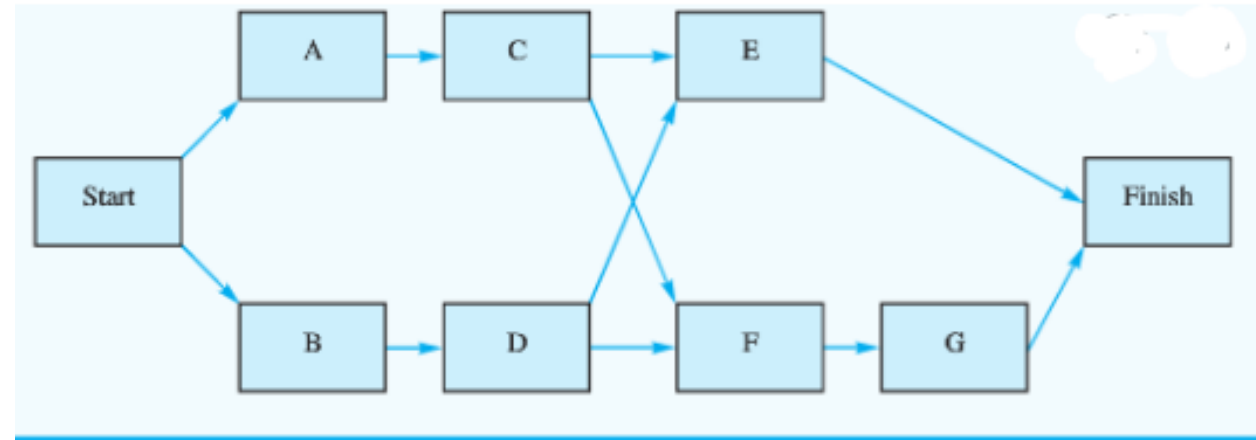


Activity	Normal Time	Crash Time	Normal Cost	Crash Cost
A	5 weeks	3 weeks	\$12,000	\$24,000
B	3 weeks	2 weeks	\$23,000	\$35,000
C	4 weeks	2 weeks	\$5,000	\$13,000
D	6 weeks	3 weeks	\$12,000	\$31,500
E	5 weeks	4 weeks	\$20,000	\$30,000
F	7 weeks	4 weeks	\$20,000	\$42,500
G	9 weeks	5 weeks	\$25,000	\$48,000
H	8 weeks	7 weeks	\$10,000	\$18,000



Tutorial 5. (Time Cost Trade Off)

The net work schedule and necessary data of a construction project are given. Compute the Optimum duration and least cost schedule. The overhead of the project is \$650.



Activity	A	B	C	D	E	F	G
Time	3	2	5	5	6	2	2

The crashing data for this project are as follows:

Activity	Time (days)		Cost (\$)	
	Normal	Crash	Normal	Crash
A	3	2	800	1400
B	2	1	1200	1900
C	5	3	2000	2800
D	5	3	1500	2300
E	6	4	1800	2800
F	2	1	600	1000
G	2	1	500	1000